

2024 MnDOT Mobile Source Data for AADT Implementation Analysis

Introduction

Since 2021, the MnDOT Traffic Forecasting and Analysis (TFA) unit has been following an FHWA-approved implementation plan to use probe-based vehicle AADT estimates when statistical review supports their use. The plan indicated that in subsequent years, MnDOT would test whether mobile source estimate accuracy was suitable for lower-volume roadways. Last year, we evaluated mobile source estimate accuracy in two volume ranges—10,000–19,999 and $\geq 20,000$ —and found accuracy and precision estimates to be acceptable. This year, we have expanded our evaluation to include an additional lower-volume range: 7,500–9,999.

Methodology

TFA collected data from all continuous counter locations with a past AADT estimate of 7,500 or greater. To ensure sufficient ground truth data for roadways in the 7,500–9,999 AADT group, we also sampled AADTs from factored 48-hour short-term count locations within this range.

StreetLight uses OpenStreetMap (OSM) as its base network. In early March 2025 StreetLight provided TFA with the 2024 estimated AADTs for Minnesota based on these OSM segments. MnDOT staff spatially matched count locations with a past AADT of 7,500 or above from the MnDOT Linear Referencing System (LRS) to the corresponding increasing and decreasing OSM roadway segments from StreetLight. This resulting dataset of Annual Average Daily Traffic (AADT) estimates was used in TFA's statistical review.

As a quality control measure, MnDOT applies a percent tolerance criterion against all factored short-term counts, specifying the acceptable percent AADT change from a previous count. For sites with a past official AADT range of 5,000–49,999, the acceptable change is $\pm 15\%$; for volumes of 50,000+, the acceptable change is $\pm 10\%$. If the adjusted volume exceeds the acceptable percent change margin, a recount is requested. All probe-based estimates at continuous count locations were also subjected to this criterion and rejected if outside of tolerance. This criterion was also applied to estimates at short-term count locations in the 7,500–9,999 group with a prior count in 2023.

In all, 662 count locations were selected for comparison. AADT at these locations was calculated using the AASHTO AADT estimation method employed by our traffic data management software, Jackalope (High Desert Traffic LLC, NM). For each volume range, we computed the mean absolute percent error (MAPE), the standard deviation of the MAPE, and the signed percent error.

Results

In the 20,000+ group, the mean absolute percent error (MAPE) between estimates was 5.81%, with a standard deviation of 4.42%. In the 10,000–19,999 group, the MAPE was 6.28%, with a standard deviation of 4.98%. In the 7,500–9,999 group, the MAPE was 12.2%, with a standard deviation of 8.30% (Table 1).

For each of the three groups, plotting the probe-based estimates against their counted/factored count AADTs shows good clustering around a line with a slope of 1 and an intercept of 0 (Figure 1). Interestingly, the MAPEs for the higher-volume groups were very similar to the 2023 estimates (2023 20,000+: 5.08%; 2023 10,000–19,999: 6.1%).

The probe-based estimates showed a slight negative bias in the 20,000+ group (-3.25%), a small positive bias (0.26%) in the 10,000–19,999 volume group, and a negative bias in the 7,500–9,999 group (-4.02%). These biases were estimated using the Wilcoxon signed-rank test, a non-parametric test that compares paired differences and tests the null hypothesis that the AADT differences have a median of zero.

The 20,000+ range had a median difference of 2603 vehicles ($p > .01$, 95% confidence interval [2,085, 3,105]), the 10,000 – 19,999 volume range had a median difference of -57 vehicles ($p = .47$, 95% confidence interval [-215, 106.5]). Finally, the 7,500 – 9,999 group had a median difference of 349.1 vehicles ($p > .01$, 95% confidence interval [161, 535]). The accuracy estimates for these volume ranges are comparable to the accuracy of traditional methods, such as the expansion of short-term counts (Publication No. FHWA-PL-21-031, page 36). Leveraging these estimates would not only improve safety for our count staff but also allow MnDOT to reallocate resources toward increasing counts on lower-volume roads, ultimately enhancing overall accuracy. Given the accuracy of these estimates and their broader benefits to our total volume program, we request approval to utilize StreetLight’s estimates for these volume ranges.

TABLE 1.

AADT Range	Mean Absolute Percent Error	Signed Absolute Percent Error	Absolute Percent Error Standard Deviation	Number of Sites
> 20,000	5.81%	-3.25%	4.42	314
10,000 - 19,999	6.28%	0.26%	4.98	157
7,500 - 9,999	12.16%	-4.02	8.30	191

FIGURE 1.

